Hazardous Celestial Objects

Final Presentation

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Data Mining Methodology I (STA5703)



SECTION 1

Problem Statement

- Predicting whether a celestial object is hazardous to Earth is critical for planetary defense.
- A direct impact could cause ecosystem disruption, food and resource shortages, and geopolitical instability.
- Misidentifying a hazardous object wastes significant national resources; missing one could be catastrophic.
- A reliable predictive model is especially important for newly discovered objects with no prior observation history.
- Our goal is simple: can we determine if a new object will pose a threat to Earth?



SECTION 2

Dataset Description

375,659	41	35,430	18
Raw Rows	Raw Columns	Final Rows	Predictors

Source: NASA NeoWs 1950–2025 via Kaggle ("Near Earth Comets Classification")

- Response variable: Hazardous. Binary categorical (TRUE / FALSE)
- Class imbalance: ~93% non-hazardous, ~7% hazardous
- Split strategy: 60% training, 20% validation, 20% test (stratified, seed = 5703)

SECTION 2

Data Cleaning & Preprocessing

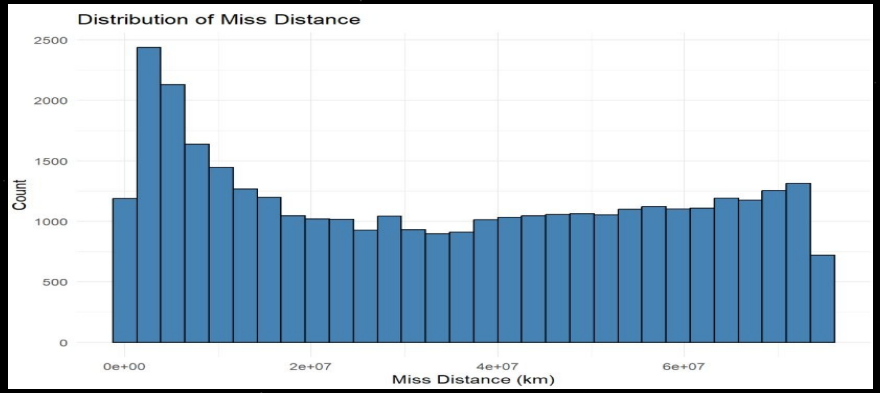
- Removed redundant columns: Name, Equinox, and Miss Distance in non-kilometer units
- Removed date and timestamp columns — no explanatory power for the model
- Filtered to Earth-orbiting objects only (dropped Jupiter, Mars, Moon, Mercury, Venus, Ceres rows)
- Removed duplicate observations based on Neo Reference ID to prevent data leakage
- Converted Minimum.Orbit.Intersection from scientific notation string to numeric
- Updated Hazardous column to factor for classification compatibility

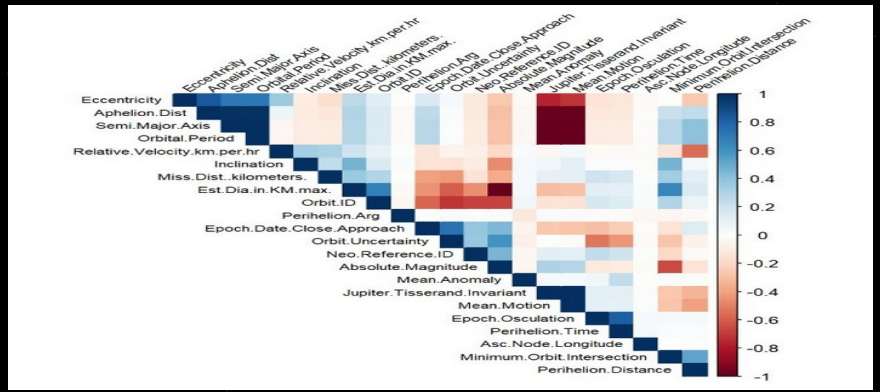


SECTION 2

Exploratory Data Analysis

- Strong class imbalance: 92.96% non-hazardous vs. 7.04% hazardous
- Miss Distance is right-skewed
  - Large magnitude range (mean ≈ 34.5M km)
- Spearman correlation used to reduce outlier influence
- Minimum Orbit Intersection emerged as the dominant predictor across all methods
- Some relationships appear nonlinear despite high correlation





SECTION 3

Best Subset & Stepwise Selection

Best Subset Selection

- Adj R^2 peaked at 0.3137 (variables 10–15)
- BIC optimal at 6 variables (BIC = −7,915.166)
- Cp minimum at 10 variables (Cp = 7.097)
- Selected 6-variable model: Absolute Magnitude, Est Dia KM Max, Orbit Uncertainty, Min Orbit Intersection, Jupiter Tisserand Invariant, Inclination
- Validation AUC: 0.9894

Stepwise Selection (Bidirectional)

- Used stepAIC() with direction = 'both'
- Baseline full model AIC: 2,240.344
- Stepwise model AIC: 2,220.360
- Selected 5 variables including Minimum Orbit Intersection and Absolute Magnitude
- Validation AUC: 0.9893

SECTION 3

Ridge & LASSO Regularization

Ridge Regression (alpha = 0)

- 5-fold cross-validation
- lambda.min = 0.008233
- lambda.1se = 0.009036
- Retains all 18 predictors — no elimination
- Validation AUC: 0.9814
- Lower AUC due to retention of noise predictors

LASSO Regression (alpha = 1)

- 5-fold cross-validation
- Performs automatic variable selection
- lambda.min selects 13 variables
- Minimum Orbit Intersection — dominant predictor
- Validation AUC: 0.9894
- Best balance of performance and simplicity

SECTION 3

Variable Selection — Model Comparison

Method	Variables	AIC	AUC (Validation)
Baseline (Full Logistic)	18	2240.344	0.9893
Stepwise (Bidirectional)	5	2220.360	0.9893
Best Subset	6	2225.745	0.9894
Ridge (lambda.min)	18	N/A	0.9814
LASSO (lambda.min)	13	N/A	0.9894

Conclusion: LASSO and best subset tied for the highest AUC (0.9894); LASSO selected because it performs variable selection automatically within cross-validation.

SECTION 4

Random Forest

Model 1 Baseline	Model 2 mtry = 4 + classwt	Model 3 Threshold 0.15	Model 4 ✓ Threshold 0.626
Sens: 99.6%	Sens: 99.8%	Sens: 100%	Sens: 99.6%
Spec: 99.8%	Spec: 99.7%	Spec: 98.8%	Spec: 99.9%
Kappa: 0.985	Kappa: 0.979	Kappa: 0.920	Kappa: 0.990
Default threshold	*Fewer FN*	*Perfect recall*	*ROC-optimized*

Test Set: Sensitivity 98.4% | Specificity 99.92% | Kappa 0.9859 | AUC 0.9993 | FN = 8 | FP = 5

SECTION 4

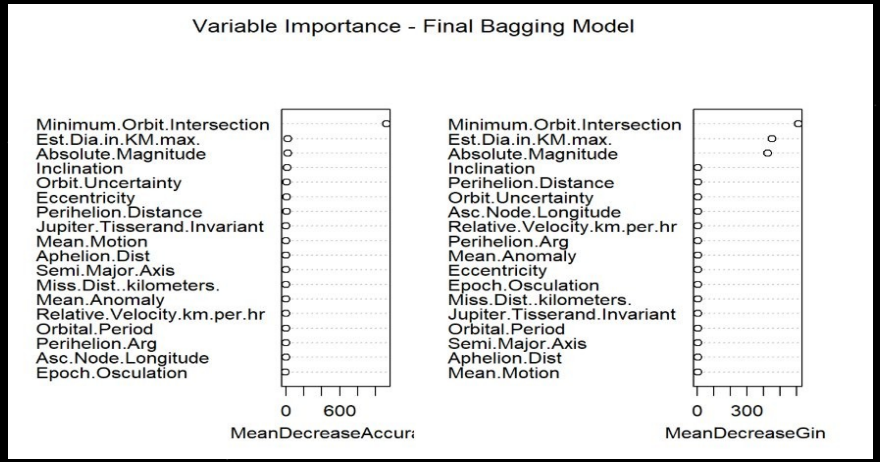
Bagging

- mtry = p (all predictors) — pure bagging
- Trained 100, 300, and 500 tree models
- OOB error stabilized around 150 trees (0.17–0.19%)
- Validation: Recall 100%, AUC up to 0.9999 across all models
- 300-tree model selected: best AUC with efficiency

Test Set: Sensitivity 99.6% | Specificity 96.42%
Kappa 0.7891 | AUC 0.9987 | FN = 2 | FP = 236

Validation Results

Model	Recall	AUC
Bagging_100	100%	0.9997
Bagging_300	100%	0.9999
Bagging_500	100%	0.9999



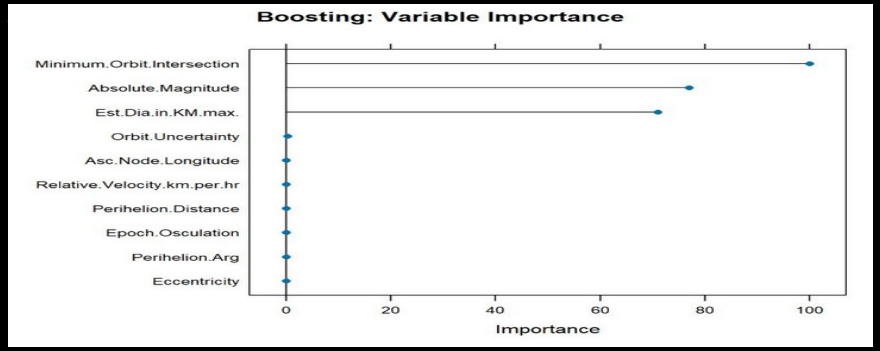
SECTION 4

Gradient Boosting

- GBM via caret
  - 5-fold cross-validation with downsampling
- Grid search: Trees 100–600, Depth 1–5, Shrinkage 0.001/0.01/0.1
- Optimal: 500 trees, depth = 5, shrinkage = 0.01, min obs = 20
- ROC-optimized classification threshold: 0.929
- Top predictors: Minimum Orbit Intersection, Absolute Magnitude, Est Dia KM Max

Validation Set

Metric	Value
Accuracy	99.89%
Sensitivity	99.60%
Specificity	99.91%
Kappa	0.9914
AUC	0.9999
False Negatives	2



Test Set: Sensitivity 98.4% | Specificity 99.91%
Kappa 0.9849 | AUC 0.9996 | FN = 8 | FP = 6

SECTION 5

Model Performance Comparison

Model	Accuracy	Sensitivity	Specificity	Kappa	AUC	FN
Random Forest	**0.9982**	**0.9840**	**0.9992**	**0.9859**	**0.9993**	**8**
Bagging	0.9664	0.9960	0.9642	0.7891	0.9987	2
Boosting	**0.9980**	**0.9840**	**0.9991**	**0.9849**	**0.9996**	**8**

Random Forest and Boosting tied: 8 FN each, with 5 vs. 6 false positives. Bagging missed only 2 hazardous objects, at the cost of 236 false positives.

SECTION 6

Conclusion

- All three ensemble models demonstrated extremely strong performance with AUC > 0.998.
- Random Forest and Boosting tied for the best overall balance –
  - Both missed 8 of 499 hazardous objects. RF had 5 FP (Kappa 0.9859); Boosting had 6 FP and the highest AUC (0.9996).
- Bagging achieved the highest sensitivity (99.6%) with only 2 FN using threshold 0.025, but at the cost of 236 false positives.
- The best choice depends on the cost of a missed object versus a false alarm.
- Ensemble methods show clear improvement over previous LASSO model (AUC = 0.9894).
- Key predictors consistent across all methods: Minimum Orbit Intersection, Absolute Magnitude, Est. Diameter (km max).
- Future work: additional feature engineering, newly discovered near-Earth objects, non-Earth-orbiting hazardous objects.

Thank You

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Questions?

